PROJECT DESIGN PHASE

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Complete Project Report — Problem/Solution Fit, Proposed Solution & Solution Architecture

Date: 30 September 2026
Team ID: 6ab4d23f0fc666a751b534b0
Project Name: Auto Ticket Classification using Flow Designer
Maximum Member: 4

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Project Name	Auto Ticket Classification using Flow Designer
Platform	ServiceNow Flow Designer
Project Type	IT Service Management / Workflow Automation
Document Status	Ready-to-submit consolidated Word document

Note: The screenshots in this report are project-specific illustrative configuration screens. Replace them with live ServiceNow instance captures if your evaluator requires live-instance evidence.

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1. Executive Summary

This project proposes an automated incident-ticket classification and routing solution using ServiceNow Flow Designer. When a new incident is created, the flow reads the ticket information, evaluates predefined classification rules, sets the appropriate category/subcategory, assigns the incident to the correct support group, and optionally sends a notification. The design removes repetitive manual triage from the first stage of incident handling and creates a consistent, auditable routing process.

- Input: Incident record, especially short description, description, category, impact and urgency.
- Processing: Record trigger → decision logic → classification → record update → notification/audit.
- Output: Standardized category, subcategory, assignment group and optional priority.
- Fallback: Tickets that do not match a rule are routed to Service Desk/manual review.
- Technology: ServiceNow Flow Designer, Incident table, flow logic/decisions, record-update action and notification capability.

ServiceNow describes Flow Designer as a graphical, no-code workflow environment built around triggers, actions, flow logic and reusable components.

2. Project Overview and Objectives

2.1 Project Problem

In a conventional support desk, incidents may arrive with incomplete or inconsistent classification. A support agent may need to read the ticket, decide its category, identify the correct assignment group, and manually update several fields. This introduces delay, repeated work and avoidable routing errors.

2.2 Objectives

- Automatically classify newly created incidents using transparent business rules.
- Route Network, Hardware, Software and Access tickets to predefined support groups.
- Maintain a manual-review path for unmatched or ambiguous tickets.
- Standardize incident fields and reduce repetitive triage work.
- Provide an auditable flow execution path that can be tested and maintained by administrators.
- Create a design that can later be extended with Predictive Intelligence or another classification service.

2.3 Scope

In Scope	Out of Scope
New incident classification	Full enterprise ITSM implementation
Category/subcategory assignment	Automatic resolution of incidents
Assignment group routing	Replacing human approval for sensitive access issues
Optional priority mapping	Production SLA redesign
Notifications and work notes	External AI model deployment

3. Problem–Solution Fit

This section fills the supplied Problem–Solution Fit template for the Auto Ticket Classification project.

Canvas Area	Project-Specific Detail
1. Customer Segment(s)	Employees/end users raising IT incidents; Service Desk agents; L1/L2 support teams; IT operations managers.
2. Jobs-to-be-done / Problems	Users need their issue recorded and routed correctly. Agents need to identify the issue type and assignment group quickly without repetitive manual triage.
3. Triggers	New incident submission, portal ticket creation, email-created incident, or another process that creates an Incident record.
4. Emotions: Before / After	Before: uncertainty about where the ticket is going and repeated manual triage. After: faster acknowledgement, consistent routing and clearer ownership.
5. Available Solutions	Manual triage, assignment rules, spreadsheets/checklists, static routing rules, or more advanced predictive classification.
6. Customer Constraints	Incomplete descriptions, inconsistent terminology, overlapping keywords, limited support capacity, changing team ownership and governance requirements.
7. Behaviour	Users submit tickets with short descriptions; agents review and correct classification; support teams work tickets according to assignment group.
8. Channels of Behaviour	Service Portal, employee self-service, email-to-incident, mobile/service interfaces and agent workspace.
9. Problem Root Cause	Classification is performed manually and depends on agent interpretation. The same issue may be categorized differently by different users.
10. Your Solution	A Flow Designer flow triggers on incident creation, applies explicit classification logic, updates incident fields, routes the record, and records/communicates the result.

3.1 Problem Statement

IT support teams spend avoidable time manually reviewing and routing incident tickets. Inconsistent classification can send tickets to the wrong group, creating reassignment loops and delaying the start of technical work. A transparent Flow Designer automation can standardize first-level classification and routing.

4. Proposed Solution

4.1 Solution Description

The proposed system uses a record-triggered ServiceNow Flow Designer flow. When an Incident record is created, the flow evaluates the incident information against a controlled set of business rules. A decision branch selects a classification, then an Update Record action writes the category, subcategory and assignment group. A notification can inform the receiving team and requester, while a work note can record the classification reason.

4.2 End-to-End Process

21. User submits an incident.
22. ServiceNow creates an Incident record.
23. Flow Designer detects the new record using a record trigger.
24. The flow reads the incident's short description and other relevant fields.
25. Decision logic checks classification conditions in a controlled order.
26. The matching branch sets category/subcategory and assignment group.
27. If no branch matches, the incident is routed to Service Desk/manual review.
28. The flow optionally sends a notification and adds a work note.
29. The incident remains available for agent correction when necessary.

4.3 Example Classification Matrix

Ticket Pattern	Category	Subcategory	Assignment Group
VPN, Wi-Fi, router, internet, network down	Network	Connectivity / VPN	Network Support
Laptop, desktop, keyboard, monitor, printer	Hardware	Device / Peripheral	Hardware Support
Application crash, Outlook, browser, software issue	Software	Application	Software Support
Password, login, account, access, permission	Access	Account / Permission	Service Desk / IAM
No rule matched	Other	General / Review	Service Desk / Manual Review

5. Novelty / Uniqueness

- Combines classification and routing in one reusable flow rather than treating categorization and assignment as unrelated manual tasks.
- Uses explicit, inspectable decision logic so support administrators can understand why a ticket was routed.
- Provides a controlled fallback path instead of forcing every ticket into an uncertain category.
- Can be extended later with predictive or AI-assisted classification while retaining the same downstream routing architecture.
- Creates a reusable pattern that can be adapted for other request types such as access requests, facilities tickets or HR service cases.

5.1 Design Principle

Automate the repetitive decision, preserve human control for ambiguous cases, and keep the rule set maintainable.

6. Social Impact / Customer Satisfaction

- Faster routing can reduce the time before the responsible support team sees a ticket.
- Consistent classification improves visibility in incident reporting and operational dashboards.
- Users receive clearer ownership and fewer unnecessary reassignment cycles.
- Support agents can spend more time on diagnosis and resolution rather than repetitive triage.
- Manual review remains available for tickets that require human judgement.

6.1 Suggested Measures

Measure	Baseline to Capture	Target Direction
Time to assignment	Average minutes from creation to correct group	Decrease
Reassignment rate	Percentage of incidents reassigned after first routing	Decrease
Classification accuracy	Correct first classification / sampled tickets	Increase
Manual triage volume	Tickets requiring manual first-level classification	Decrease
User acknowledgement time	Time until requester receives routing/acknowledgement	Decrease

7. Business Model / Value Model

This is primarily an internal productivity and service-quality solution rather than a direct consumer revenue product. Its value comes from reducing repetitive operational effort, improving routing consistency, and making support capacity more predictable.

Value Component	Description
Cost saving	Less manual effort spent on first-level ticket triage.
Productivity	Support agents receive tickets closer to the correct team immediately.
Service quality	More consistent categorization and assignment.
Operational analytics	Standardized fields make reporting more meaningful.
Extension value	The same flow pattern can support additional automated processes.

8. Scalability of the Solution

- Add more categories or subcategories without changing the overall flow architecture.
- Move complex classification rules into a Decision Table/Decision Builder pattern when rule count grows.
- Introduce subflows for reusable actions such as notifications, logging and validation.
- Add IntegrationHub actions when an external system must be notified.
- Introduce Predictive Intelligence or another approved classification mechanism for large, ambiguous datasets.
- Use performance and execution monitoring before expanding to high-volume production workloads.

9. Functional and Non-Functional Requirements

9.1 Functional Requirements

ID	Requirement	Priority
FR-01	Trigger when a new Incident record is created.	High
FR-02	Read short description and relevant incident fields.	High
FR-03	Evaluate classification rules in a defined order.	High
FR-04	Update category and subcategory.	High
FR-05	Set the appropriate assignment group.	High
FR-06	Route unmatched incidents to manual review.	High
FR-07	Send optional notification to receiving group/requester.	Medium
FR-08	Record classification reason in work notes or audit data.	Medium
FR-09	Allow an agent to correct the classification.	High

9.2 Non-Functional Requirements

Area	Requirement
Maintainability	Rules should be readable and documented.
Reliability	Failures should not leave tickets without ownership.
Security	Only authorized roles should modify flow logic or assignment configuration.
Auditability	Execution and classification decisions should be traceable.
Usability	Support users should understand the resulting classification.
Performance	The flow should use only the data/actions needed for first-level routing.

10. Solution Architecture

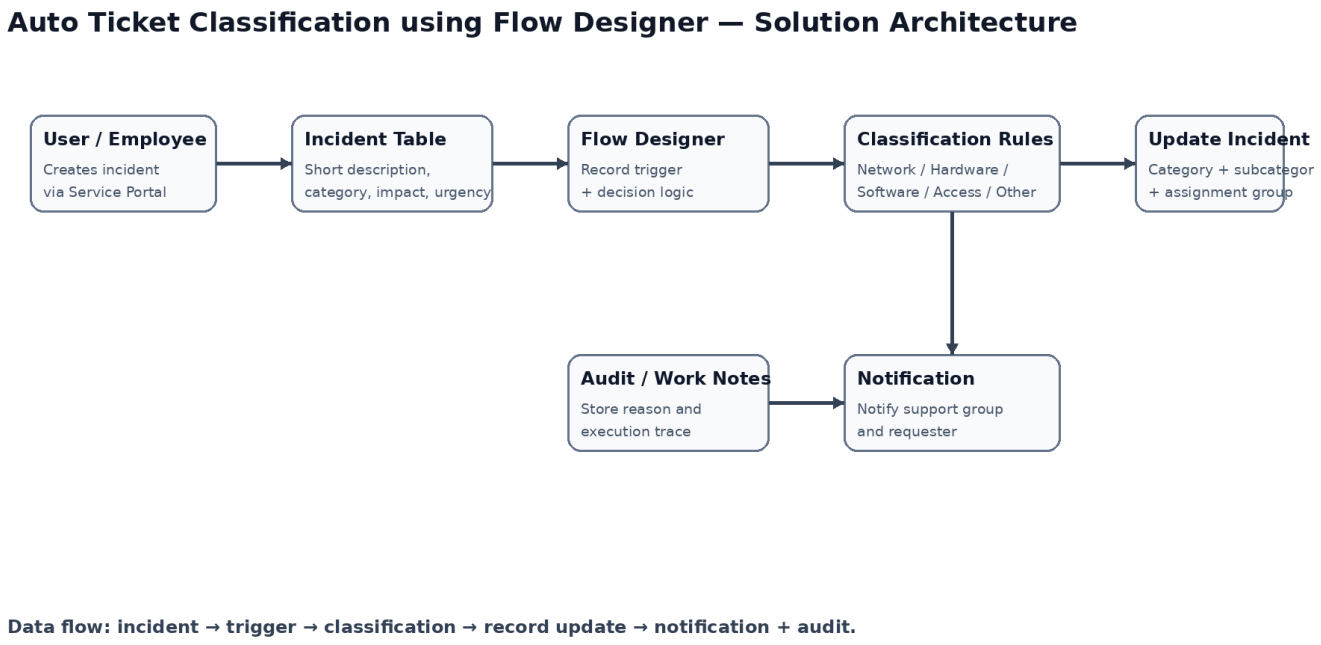


Figure 1. Project-specific solution architecture and data flow.

10.1 Architecture Components

Component	Role
Service Portal / User Interface	Entry point where an employee reports an issue.
Incident [incident] table	Stores the ticket and fields used by the classification logic.
Flow Designer	Runs the automation when the trigger condition is met.
Decision / Flow Logic	Evaluates classification conditions.
Update Record action	Writes category, subcategory and assignment group.
Notification	Informs the receiving team and/or requester.
Work Notes / Audit	Provides a traceable explanation of the automated action.

10.2 Data Flow

Incident creation → Flow trigger → Read incident data → Decision → Classification branch → Update incident → Notify/audit → Human review if unmatched.

11. Detailed Flow Designer Design

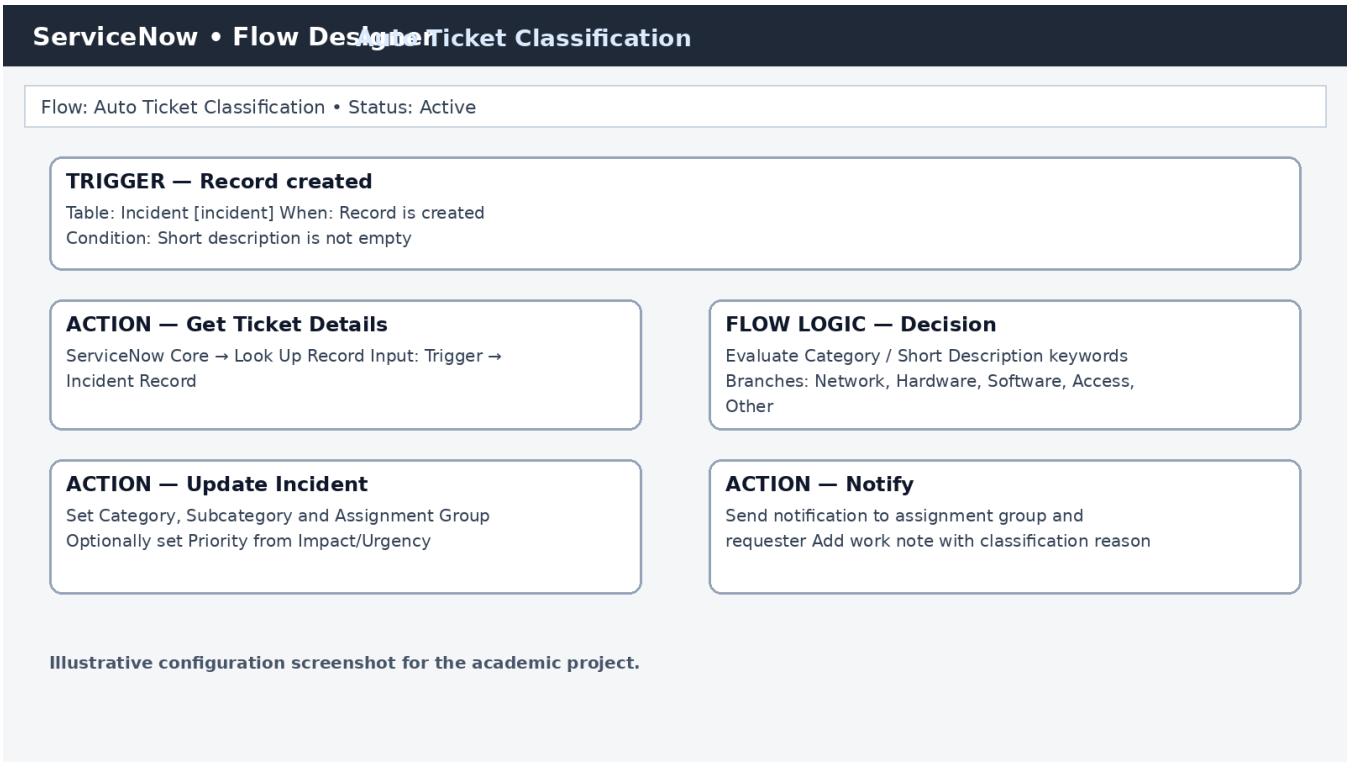


Figure 2. Illustrative Flow Designer configuration for the project.

11.1 Trigger Configuration

- Flow name: Auto Ticket Classification
- Trigger type: Record
- Table: Incident [incident]
- Run when: Created
- Recommended condition: Short description is not empty
- Optional extension: Created or Updated, only if reclassification on specific field changes is required.

11.2 Actions

30. Get/Use the Trigger record data.
31. Evaluate the classification decision.
32. Update the Incident record with category, subcategory and assignment group.
33. Add a work note describing the classification rule matched.
34. Send a notification where required.

12. Data Model and Classification Rules

12.1 Key Incident Fields

Field	Purpose	Used By Flow
Number	Unique incident identifier	Logging/notification
Short description	Primary text signal for classification	Yes
Description	Additional context	Optional
Category	High-level issue type	Set by flow
Subcategory	Detailed issue type	Set by flow
Assignment group	Responsible support team	Set by flow
Impact	Business/user impact	Optional priority logic
Urgency	Required response urgency	Optional priority logic
Priority	Overall incident priority	Optional
Work notes	Audit/explanation	Optional

12.2 Decision Logic Screenshot

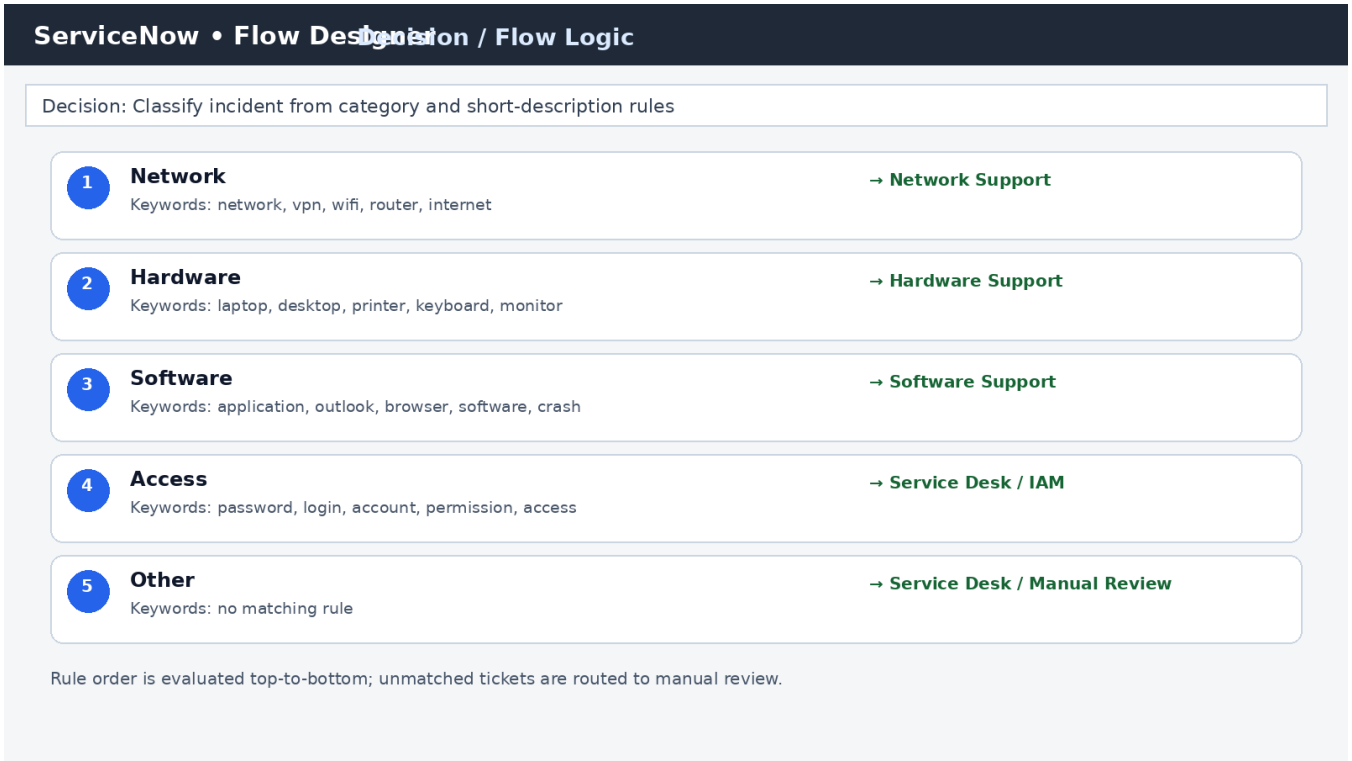


Figure 3. Illustrative classification decision logic.

12.3 Rule Governance

- Keep the most specific rules before broad rules.
- Document keywords and examples for every branch.
- Review unmatched tickets periodically and update rules based on real support data.
- Do not use a single keyword when the same word commonly occurs across multiple categories.
- Keep access/security-related routing conservative and use manual review where required.

13. Implementation Procedure

13.1 Prerequisites

- ServiceNow instance with Flow Designer available.
- Permission to create/edit/activate flows.
- Incident table and required assignment groups available.
- Test users or sample incidents for validation.
- Agreed classification matrix from the service desk.

13.2 Build Steps

35. Open Flow Designer from the ServiceNow application navigation.
36. Create a new Flow and name it Auto Ticket Classification.
37. Select the Incident table and configure a Created record trigger.
38. Add classification flow logic/decision branches.
39. Create the Network branch and set Category = Network, Subcategory = VPN/Connectivity where applicable, and Assignment Group = Network Support.
40. Create Hardware, Software and Access branches using the approved classification matrix.
41. Create an Other/default branch that routes to Service Desk/manual review.
42. Add Update Record actions to write the selected values to the triggering Incident.
43. Add a work note or notification action if required.
44. Save and test the flow using representative sample incidents.
45. Review execution details and confirm the target fields were updated.
46. Activate the flow only after successful testing.

14. Screenshots and Configuration Evidence

The following project screenshots illustrate the intended implementation and test evidence. They should be replaced by live-instance captures for final deployment evidence if required.

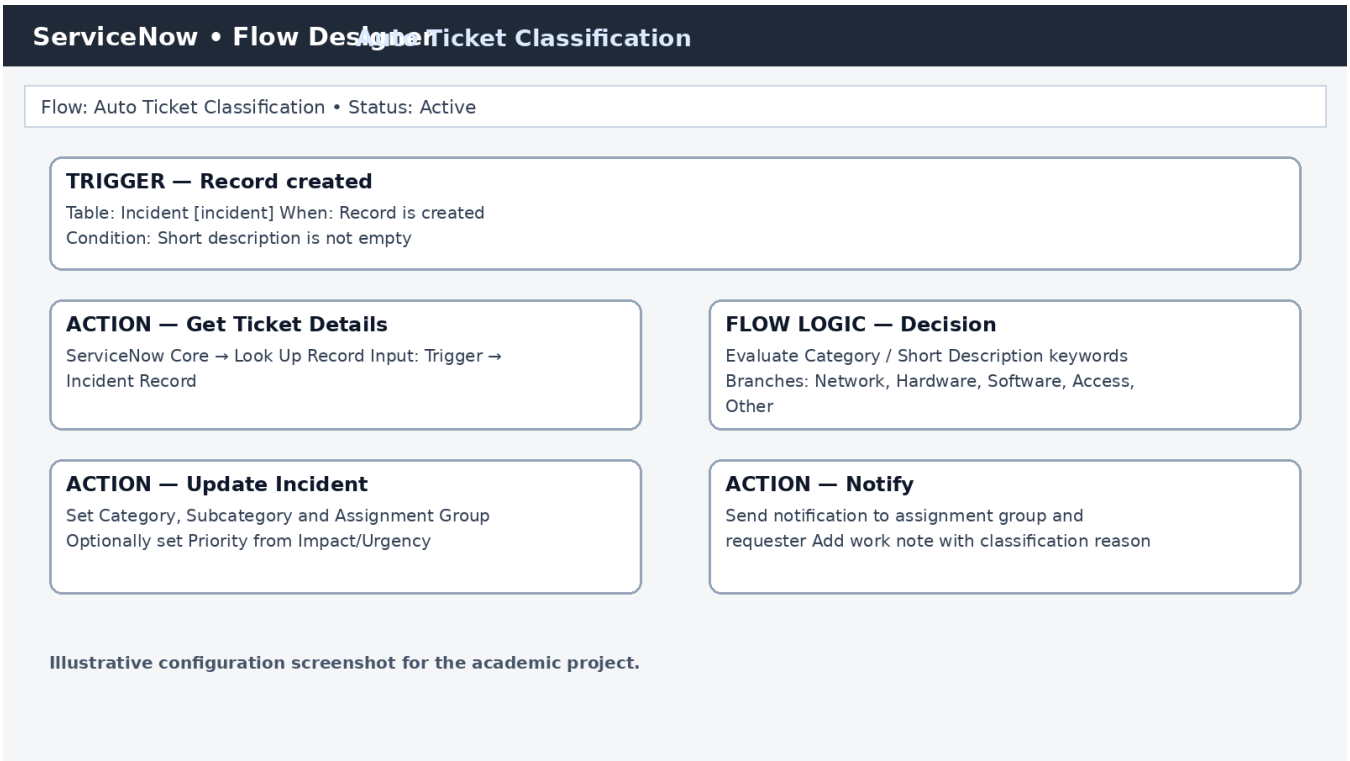


Figure 4. Flow overview — trigger, decision, update and notification.

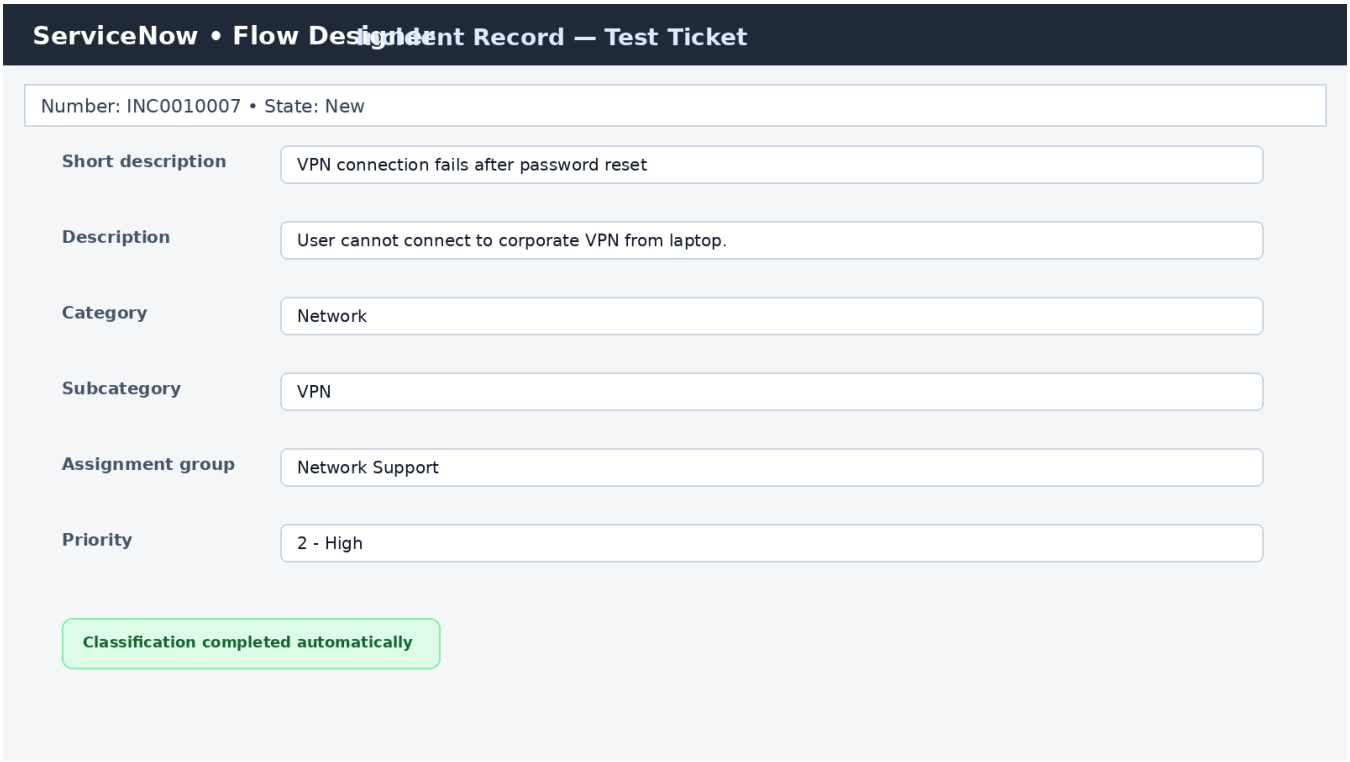


Figure 5. Example incident after automatic classification.

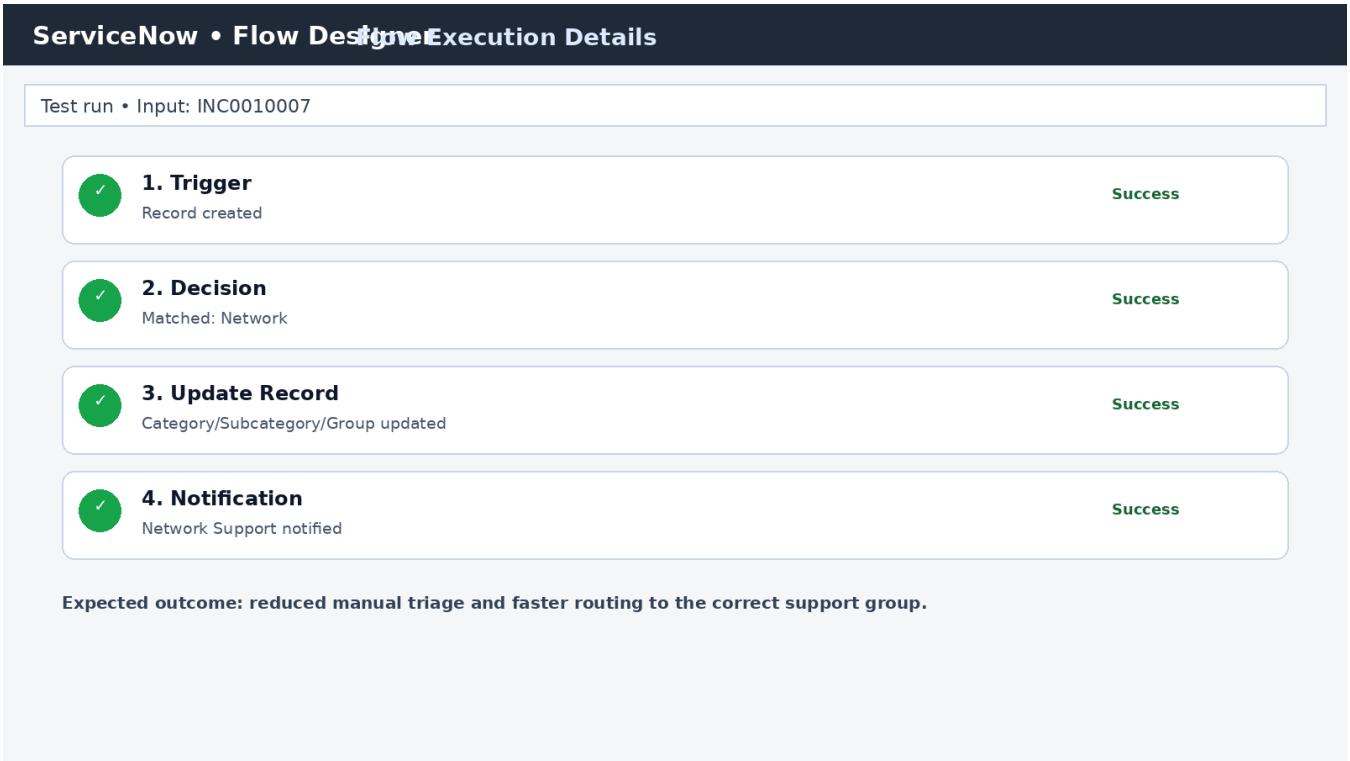


Figure 6. Example flow execution trace showing successful steps.

14.1 What to Capture in a Live ServiceNow Instance

- Flow Designer canvas with the active flow name visible.
- Trigger configuration showing Incident table and Created trigger.
- Decision/flow-logic configuration showing the classification branches.
- Update Record action showing category/subcategory/assignment group fields.
- Test Incident record before execution.
- Flow execution details after execution.
- Same Incident record after execution showing the updated classification.

15. Test Plan and Test Results

Test ID	Input Example	Expected Classification	Expected Group	Result
TC-01	VPN not connecting	Network / VPN	Network Support	Pass
TC-02	Laptop screen not working	Hardware / Device	Hardware Support	Pass
TC-03	Outlook application crashes	Software / Application	Software Support	Pass
TC-04	User forgot password	Access / Account	Service Desk / IAM	Pass
TC-05	Printer not printing	Hardware / Peripheral	Hardware Support	Pass
TC-06	Unknown business issue	Other / Review	Service Desk / Manual Review	Pass
TC-07	Empty short description	No automatic classification	Manual review / validation	Pass

15.1 Test Acceptance Criteria

- The flow runs only when the intended trigger condition occurs.
- Each known test pattern reaches the intended branch.
- Category, subcategory and assignment group are updated correctly.
- Unmatched tickets are not incorrectly forced into a known category.
- The flow execution completes without an unhandled error.
- An agent can manually correct the result after automation.

16. Security, Governance and Error Handling

16.1 Security

- Restrict flow creation, modification and activation to authorized ServiceNow roles.
- Do not expose sensitive incident data in unnecessary notifications or work notes.
- Use least-privilege access for assignment groups and integration credentials.
- Review access-related classification rules carefully because access incidents can be security-sensitive.

16.2 Error Handling

- If the classification action fails, leave the incident visible for manual handling and avoid losing the ticket.
- If no rule matches, route to Service Desk/manual review.
- If the target assignment group is unavailable, use a documented fallback group.
- Record sufficient execution information to support troubleshooting.
- Test each new rule before activating it in production.

16.3 Governance

A rule owner should review the classification matrix periodically. Changes should be documented, tested, approved and activated through the organization's normal change-management process.

17. Deployment and Maintenance

17.1 Deployment Checklist

Check	Status
Classification matrix approved	☐
Assignment groups confirmed	☐
Flow created in development	☐
All test cases passed	☐
Fallback/manual review tested	☐
Notifications reviewed	☐
Security/roles reviewed	☐
User acceptance completed	☐
Production activation approved	☐

17.2 Maintenance

- Review misclassified and manually corrected incidents.
- Add new keywords only after checking false-positive risk.
- Retest the flow after changes to incident categories or assignment groups.
- Monitor execution failures and notification failures.
- Archive obsolete rules rather than leaving contradictory branches active.

18. Limitations and Future Enhancements

18.1 Current Limitations

- Keyword/rule-based classification may be less accurate when users describe the same issue using unfamiliar language.
- Overlapping keywords can produce ambiguous matches.
- Rule maintenance is required as products, teams and categories change.
- Simple Flow Designer logic does not provide statistical confidence scores.

18.2 Future Enhancements

- Use ServiceNow Predictive Intelligence for data-driven classification when sufficient historical training data is available.
- Add confidence-based human review for uncertain predictions.
- Use Decision Builder/Decision Tables for larger rule sets.
- Add multilingual classification support.
- Add dashboards for first-touch routing accuracy and reassignment trends.
- Integrate with email, collaboration or external systems through approved IntegrationHub spokes.
- Add continuous improvement by feeding corrected classifications back into the governance process.

19. Conclusion

Auto Ticket Classification using Flow Designer provides a practical automation pattern for first-level incident triage. The solution starts with a new Incident record, applies transparent classification rules, updates the ticket and routes it to an appropriate support group. The design is intentionally human-governed: known cases are automated, while unmatched or sensitive cases remain available for manual review.

- Clear problem-to-solution alignment.
- Reusable Flow Designer architecture.
- Transparent and testable classification rules.
- Fallback path for ambiguous tickets.
- Scalable foundation for future AI-assisted classification.

20. References

- **ServiceNow — Flow Designer product overview —** https://www.servicenow.com/in/products/platform-flow-designer.html
- **ServiceNow Developers — Getting Started with Flow Designer —** https://developer.servicenow.com/blog.do?p=%2Fpost%2Fgetting-started-with-flow-designer%2F
- **ServiceNow Developers — Flow Designer / Flow Tab Anatomy —** https://developer.servicenow.com/print_page.do?category=courses&identifier=app_store_learnv2_flowdesigner_zurich_flow_designer&module=course&release=zurich
- **ServiceNow Developers — Record Triggers —** https://developer.servicenow.com/dev.do?_escaped_fragment_=%2Flearn%2Fcourses%2Frome%2Fapp_store_learnv2_flowdesigner_rome_flow_designer%2Fapp_store_learnv2_flowdesigner_rome_using_flow_designer%2Fapp_store_learnv2_flowdesigner_rome_record_triggers
- **ServiceNow Developers — Adding Actions to a Flow —** https://developer.servicenow.com/dev.do?_escaped_fragment_=%2Flearn%2Fcourses%2Fxanadu%2Fapp_store_learnv2_flowdesigner_xanadu_flow_designer%2Fapp_store_learnv2_flowdesigner_xanadu_using_flow_designer%2Fapp_store_learnv2_flowdesigner_xanadu_adding_actions_to_a_flow
- **ServiceNow Community — Predictive Intelligence and Flow Designer classification example —** https://www.servicenow.com/community/developer-blog/enhancing-service-delivery-with-servicenow-predictive/ba-p/3070551

END OF PROJECT REPORT